Medical Image Classification for Early Melanoma Detection

Bryan Torres
Virginia Tech
Department of Computer Science
bryantorres@vt.edu

Roman Chenoweth
Virginia Tech
Department of Computer Science
romanc22@vt.edu

Cian Teague
Virginia Tech
Department of Computer Science
cianteague@vt.edu

Abstract

Melanoma is a high-risk skin cancer for which early detection can substantially improve outcomes. Our capstone project studies binary classification of dermoscopic lesion images into benign and malignant categories using transfer-learned convolutional neural networks. This milestone report captures the first substantial phase of the project: dataset preparation, exploratory analysis, augmentation design, training infrastructure, and an initial benchmark across five pretrained architectures. Using an ISIC-derived Kaggle dataset with 13,879 total images, we implemented a stratified train/validation split, a standardized augmentation pipeline (flips, full-rotation, color jitter, MixUp, and CutOut), and a two-phase fine-tuning procedure. Preliminary results on the balanced held-out test set show strong performance across all models, with ResNet-50 currently achieving the best AUC-ROC of 0.9800, accuracy of 0.9185, recall of 0.8820, and sensitivity of 0.9850 at 80% specificity.

1 Introduction

1.1 Problem Overview

This project studies whether modern deep neural networks can help distinguish malignant melanoma from benign skin lesions in dermoscopic images. The motivating use case is not autonomous diagnosis, but decision support, so that suspicious lesions are not missed, and to avoid false alarms. This makes melanoma screening a machine learning problem with asymmetric error costs, where false negatives are more consequential than false positives.

From a technical perspective, the project is a supervised binary image classification task with three major challenges. First, dermoscopic images vary widely in lesion size, illumination, zoom, and artifact content such as rulers, hair, and dark borders. Second, medical-image performance must be evaluated with clinically interpretable metrics rather than accuracy alone. Third, benchmark quality depends heavily on reliable data splitting and preprocessing, since small implementation errors can produce misleading validation results.

1.2 Formal Problem Definition

Let an image be represented by

$$x \in \mathbb{R}^{224\times224\times3},$$

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and let the label be

$$y \in \{0, 1\},$$

where $y = 0$ denotes benign and $y = 1$ denotes malignant. We seek a model

$$f_\theta(x) \in [0, 1]$$

that estimates the probability that a lesion is malignant. The final class prediction at threshold τ is

$$\hat{y} = \mathbb{I}[f_\theta(x) \geq \tau].$$

The project therefore becomes the problem of learning parameters θ that maximize discrimination between benign and malignant lesions while emphasizing clinically important operating points such as sensitivity at fixed specificity.

2 Response to Proposal Feedback

Comment: "Maybe in the results section, add a specific subsection analyzing how different focal loss gamma parameters impact performance."

Response: We incorporated this suggestion by adding a dedicated results subsection on focal loss sensitivity to the focusing parameter γ. Because focal loss was already part of our planned imbalance-mitigation ablation, this milestone now makes that analysis more explicit by isolating the effect of different γ values rather than only mentioning focal loss as one candidate strategy.

3 Related Work

Deep learning for skin lesion analysis is well established, but prior work differs in dataset scale, clinical framing, architecture choice, and treatment of class imbalance.

Esteva et al. (1) showed that transfer learning can achieve dermatologist-level performance on curated skin lesion datasets using a CNN trained end-to-end. Their work established the core paradigm adopted in our project: initialize from ImageNet and fine-tune for lesion classification.

Haenssle et al. (2) compared a deep CNN against 58 dermatologists on dermoscopic melanoma recognition. A major strength of that study is its clinically grounded framing, since it reports model behavior at matched sensitivity/specificity thresholds. That perspective strongly influenced our evaluation design, especially our use of sensitivity at 80% and 90% specificity.

The HAM10000 dataset curated by Tschandl et al. (3) and the ISIC challenge benchmark summarized by Codella et al. (4) gave the field a reproducible experimental foundation. These resources standardized training and evaluation, but they also exposed important limitations: class imbalance, dataset bias, duplicate-like samples, and limited diversity across imaging conditions and skin tones.

Among model backbones, ResNet (5), DenseNet (6), EfficientNet (7), and MobileNetV2 (8) offer different trade-offs between representation power, parameter count, and inference cost. Rather than committing to a single backbone early, we compare all four families under the same augmentation and training protocol.

For imbalance mitigation, focal loss (9) provides a differentiable alternative to plain class weighting by emphasizing hard examples. For regularization, MixUp (10) and CutOut (11) are lightweight, architecture-agnostic augmentations that can improve generalization and calibration. Our implementation is heavily influenced on these three ideas, further expanding later in our final report.

Overall, the literature suggests three lessons that shaped our milestone design: use transfer learning, evaluate beyond accuracy, and separate architectural comparisons from data-handling decisions. The current milestone report is the first concrete execution of that plan.

4 Data Collection Process and Initial Dataset Analysis

4.1 Dataset Source

We use the "Skin Cancer: Malignant vs. Benign" dataset derived from the ISIC archive and hosted on Kaggle. The dataset was provided to us as a directory-structured image collection rather than as raw

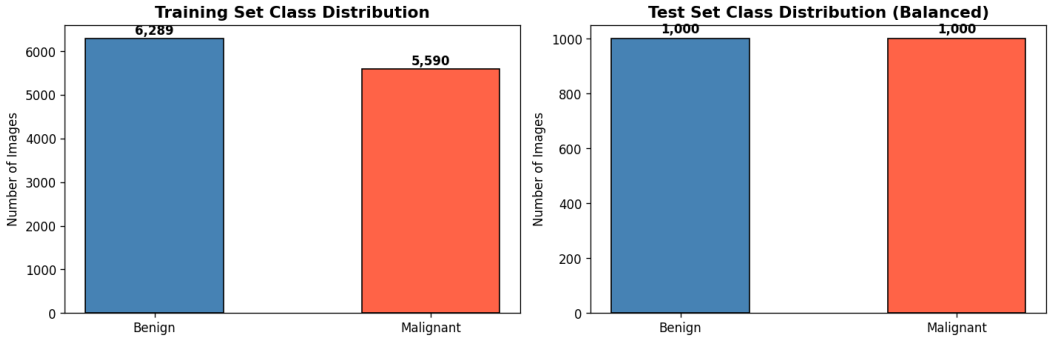

Figure 1: Class counts in the packaged dataset. The provided training directory is only mildly imbalanced, while the test set is perfectly balanced.

clinical records. Our work therefore focuses on responsible curation, preprocessing, and evaluation of the image data we were given, not on collecting new patient data.

All files are RGB dermoscopic images stored in JPEG format. During training and evaluation, every image is resized to 224×224 pixels so that it can be passed through ImageNet-pretrained architectures using a common input size.

4.2 Dataset Size and Splits

The packaged dataset contains 13,879 total images:

- **Provided training directory:** 11,879 images
- **Held-out test directory:** 2,000 images

Within the provided training directory, class counts are:

- **Benign:** 6,289 images
- **Malignant:** 5,590 images

This distribution is much less skewed than the broader class imbalance often discussed for ISIC-style datasets. In other words, the version of the dataset we received is only mildly imbalanced (approximately 1.12:1 benign-to-malignant), not severely imbalanced. After receiving the training directory, we created a **stratified** 90/10 train/validation split:

- **Training split used for fitting:** 10,692 images
  - 5,661 benign
  - 5,031 malignant
- **Validation split used for early stopping/model selection:** 1,187 images
  - 628 benign
  - 559 malignant
- **Test split:** 2,000 images (1,000 benign, 1,000 malignant)

The balanced test set is valuable because it reduces the risk that strong results are driven simply by majority-class prevalence.

4.3 Important Data Issue Discovered During Implementation

One of the most important milestone findings was not a model result but a data-pipeline issue. Our original validation split used Keras directory loading utilities. Because filenames were grouped alphabetically by class, the naive validation slice could accidentally contain almost only malignant examples. This produced pathological validation metrics and made early stopping unreliable.

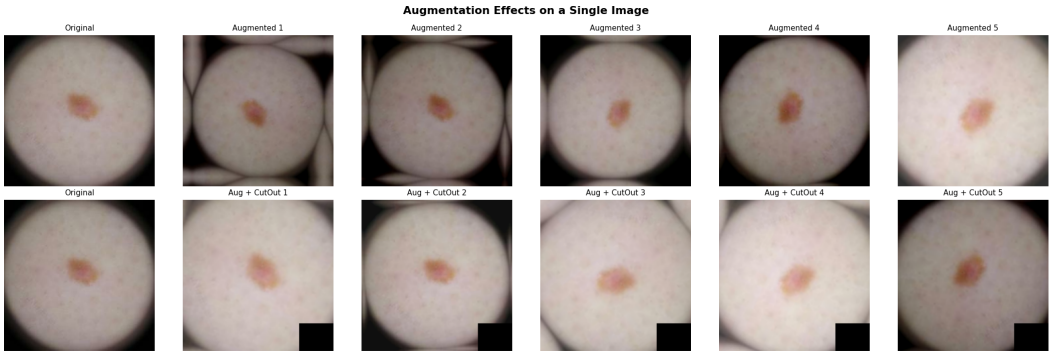

Figure 2: Examples of the augmentation pipeline applied to a single lesion image. The top row shows the original image and standard augmentations; the bottom row adds CutOut masking. MixUp is applied at the batch level during training and is therefore not shown in this figure.

To correct this, we replaced the automatic split with a manual per-class file collection and stratified shuffle split. This fix is central to the validity of all later experiments. Without it, the model-selection process would be fundamentally compromised.

5 Mathematical Background

5.1 Weighted Binary Cross-Entropy

For a predicted malignancy probability $p = f_\theta(x)$ and true label $y \in \{0, 1\}$, weighted binary cross-entropy is

$$\mathcal{L}_{\mathrm{WCE}} = -w_1 y \log p - w_0(1-y)\log(1-p),$$

where w_0 and w_1 are class weights. In our benchmark, these weights are derived from inverse class frequency in the training split. This lets us penalize the less frequent malignant class more heavily during optimization without changing the underlying dataset.

5.2 Focal Loss

One planned ablation replaces weighted cross-entropy with focal loss (9):

$$\mathcal{L}_{\mathrm{FL}} = -\alpha_t(1-p_t)^\gamma \log(p_t),$$

where p_t is the predicted probability assigned to the true class, α_t is a class-balancing coefficient, and γ is a focusing parameter. Intuitively, focal loss down-weights easy examples and concentrates learning on mistakes or uncertain cases.

5.3 MixUp and CutOut

MixUp (10) creates virtual training examples by convexly combining two images and their labels:

$$\tilde{x} = \lambda x_i + (1-\lambda)x_j, \qquad \tilde{y} = \lambda y_i + (1-\lambda)y_j,$$

where $\lambda \in [0, 1]$. In our implementation, λ is sampled from a bounded range around 0.5 to create soft labels and encourage smoother decision boundaries.

CutOut (11) masks a random square patch of the image during training. This forces the network to rely less on a single local cue and more on distributed lesion structure.

5.4 Clinical Metrics

Accuracy alone is insufficient in this setting. We therefore emphasize:

- **Sensitivity / Recall:** $\frac{TP}{TP+FN}$, the fraction of malignant lesions detected.

- **Specificity:** $\frac{TN}{TN+FP}$, the fraction of benign lesions correctly rejected.
- **AUC-ROC:** threshold-independent ranking performance over all decision thresholds.
- **Sensitivity at fixed specificity:** the maximum sensitivity achieved while meeting a chosen specificity constraint, such as 80% or 90%.
- **Brier score:** mean squared error of predicted probabilities, used as a simple measure of calibration quality.

These metrics better reflect how a model might be used in practice: not just whether it is often correct, but whether it can operate safely at clinically meaningful thresholds.

6 Implemented Methods and Algorithms

6.1 Architecture Benchmark

At this milestone stage, we completed a controlled benchmark of five pretrained architectures: EfficientNet-B0, EfficientNet-B2, ResNet-50, DenseNet-121, and MobileNetV2. Each uses the same classification head:

`GlobalAveragePooling` → `Dense(256, ReLU)` → `BatchNorm` → `Dropout(0.3)` → `Dense(1, sigmoid)`.

This shared head reduces confounding across architectures and keeps the benchmark focused on backbone quality rather than head design.

6.2 Two-Phase Fine-Tuning

Training is done in two phases:

1. **Phase 1:** freeze the pretrained backbone and train only the classification head.
2. **Phase 2:** unfreeze the backbone and fine-tune end-to-end with a lower learning rate.

Batch normalization layers remain frozen during fine-tuning to preserve stable ImageNet statistics. Early stopping and model checkpointing are driven by validation loss.

6.3 Preprocessing and Augmentation

Each architecture receives its own `preprocess_input` function from TensorFlow/Keras so that channel normalization matches the underlying pretrained backbone. Using the same augmentation stack for every architecture keeps the comparison fair. During training, all models are given the same augmentation pipeline:

- random horizontal and vertical flips,
- full-range random rotation,
- random zoom and contrast/brightness perturbations,
- CutOut,
- MixUp at the batch level.

6.4 Evaluation Procedure

Models are evaluated on the held-out balanced test set only after training is complete. Predictions are converted to probabilities, then analyzed using scikit-learn to compute ROC curves, AUC, precision, recall, F1, Brier score, and sensitivity at fixed specificity. This external metric computation is important because it avoids relying exclusively on framework-internal validation numbers.

Table 1: Initial architecture benchmark on the balanced held-out test set. All models were trained with the same preprocessing, augmentation pipeline, and weighted cross-entropy setup.

Architecture	Accuracy	AUC	Precision	Recall	F1	Sens.@80% Spec	Sens.@90% Spec	Brier
EfficientNet-B0	0.9090	0.9748	0.9495	0.8640	0.9047	0.9790	0.9480	0.0763
EfficientNet-B2	0.9090	0.9743	0.9398	0.8740	0.9057	0.9830	0.9500	0.0768
ResNet-50	**0.9185**	**0.9800**	**0.9515**	0.8820	**0.9154**	**0.9850**	**0.9600**	**0.0666**
DenseNet-121	0.9130	0.9732	0.9366	**0.8860**	0.9106	**0.9850**	0.9450	0.0748
MobileNetV2	0.9030	0.9713	0.9400	0.8610	0.8987	0.9830	0.9400	0.0808

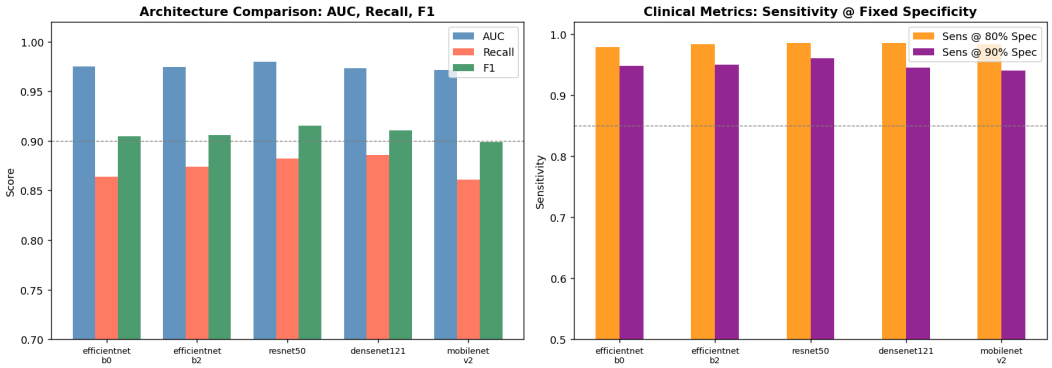

Figure 3: Preliminary comparison of AUC, recall, F1, and clinically relevant sensitivity at fixed specificity across the five benchmarked architectures.

7 Initial Findings

7.1 Architecture Benchmark Results

Table 1 reports the current benchmark results. All five architectures achieve AUC above 0.97, and all exceed 0.94 sensitivity at 90% specificity. These are encouraging early signs that the dataset is learnable under transfer learning and that the shared pipeline is functioning correctly.

7.2 Effect of the Focal Loss Focusing Parameter γ

In addition to comparing architectures, we also want to understand how focal loss changes model behavior as the focusing parameter γ varies. In our proposal, we identified focal loss as a key imbalance-mitigation strategy and planned to tune $\gamma \in \{0.5, 1, 2\}$ on the validation set. The main intuition is that larger γ values down-weight easy examples more aggressively, which can shift the model toward harder and potentially more clinically important lesion cases.

This analysis is especially relevant in our setting because melanoma screening places greater importance on avoiding false negatives than on maximizing raw accuracy alone. A small γ may behave similarly to weighted cross-entropy, while a larger γ may improve sensitivity by forcing the model to focus more strongly on ambiguous or misclassified malignant images. However, overly large values can also make optimization less stable or reduce overall calibration if the model concentrates too narrowly on a small subset of hard samples.

At the milestone stage, the full focal-loss ablation is still being completed, so we do not yet present a final table of results here. However, this subsection makes explicit that γ is not just a training detail but an interpretable hyperparameter that may materially affect recall, AUC, F1, and sensitivity at fixed specificity. In the final report, we will report a controlled comparison of $\gamma \in \{0.5, 1, 2\}$ using the same data split, augmentation pipeline, and backbone so that any observed changes can be attributed specifically to the focal-loss focusing strength.

7.3 What These Results Suggest

Several patterns are already visible:

- **ResNet-50 is the strongest all-around model so far.** It currently leads on AUC, accuracy, F1, calibration (lowest Brier score), and sensitivity at 90% specificity.
- **DenseNet-121 remains competitive for recall.** Its thresholded recall is slightly higher than ResNet-50's, which is meaningful in a clinically sensitive setting.
- **MobileNetV2 offers a deployment-friendly trade-off.** It performs slightly worse, but still reaches an AUC above 0.97 while being the lightest candidate for a later web demo.
- **The benchmark is close enough to justify deeper ablations.** Because backbone performance is strong across the board, the next gains may come more from operating-point calibration and imbalance handling than from swapping architectures again.

These are *initial* findings rather than final claims. The current report covers the first major experiment, but not yet the full set of analyses promised in the proposal.

8 General Difficulties and Lessons Learned

8.1 Validation Split Failure

The most serious issue encountered so far was an invalid validation split caused by directory-order loading. Because the malformed split produced pathological validation behavior, it could have caused incorrect early stopping decisions and misleading model comparisons. Fixing this problem was essential before any result could be trusted.

8.2 Framework Metric Reliability

A second issue came from framework-side validation metrics. In our environment, monitoring `val_auc` through TensorFlow/Keras produced unreliable behavior because stateful metric objects can be reused incorrectly across repeated `compile()` calls. We resolved this by creating fresh metric instances for every model and using validation loss for early stopping, while computing the final AUC and related metrics externally with scikit-learn.

8.3 Computation Time

Even with transfer learning, benchmarking five backbones sequentially is computationally expensive. This affects how much of the full experimental plan can be completed within a capstone timeline.

8.4 Medical-Data Limitations

Although the image labels are usable for an academic benchmark, we do not yet have a rich metadata layer for subgroup analysis, institution shift, or acquisition-device variation. As a result, we cannot make strong claims about fairness or real-world deployment readiness at this stage.

9 Remaining Work Toward the Final Report

1. **Finish the imbalance ablation study.** We have implemented weighted cross-entropy, focal loss, oversampling, and the combined focal-plus-oversampling condition, but the full experiment is still running and has not yet been summarized in the report.
2. **Complete comprehensive evaluation for the best configuration.** This includes ROC and precision-recall plots, confusion matrices, threshold analysis, and calibration curves.
3. **Run interpretability analysis.** We plan to generate Grad-CAM visualizations for both correct and incorrect predictions to see which lesion regions drive the model.
4. **Refine deployment considerations.** MobileNetV2 and the current best model will be compared for a lightweight web demonstration, likely using Streamlit.

Our current plan is to use the completed architecture benchmark to select a strong backbone (currently ResNet-50), then run the imbalance and evaluation experiments only on that reduced model set to stay within time and compute limits.

10 Conclusion

At the 30% milestone point, we now have a functioning deep learning pipeline for melanoma classification, a validated stratified split, a standardized augmentation framework, and the first major benchmark results across five transfer-learned CNN backbones. The strongest current result is a ResNet-50 model with AUC 0.9800 and sensitivity 0.9850 at 80% specificity on a balanced held-out test set. Just as importantly, we have already identified and corrected implementation problems that could have invalidated later experiments. The remaining work is now well-defined and focused: complete imbalance ablations, produce deeper evaluation and interpretability analyses, and package a small deployment demo around the best-performing configuration.

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